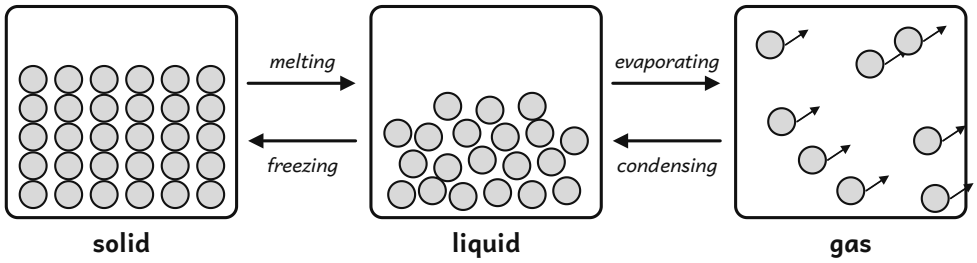
PHYSICAL AND CHEMICAL SCIENCE

# States of Matter: Evaporation, Condensation, Boiling and Melting

Solids, liquids and gases, and how heating and cooling change them




**In this pack**

📖 **Learn it:** Solids, liquids and gases: changing state ★☆☆
**Fact check:** Changes of state ★☆☆
**Think it through:** Explaining changes of state ★★☆
**Investigate:** Does surface area affect evaporation? ★★☆
**Real-world case study:** Salt from the sun: Cox's Bazar salt farmers ★★★
**Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science Stage 5: Unit 3 States of matter (3.1–3.9).

**This pack belongs to:** _____ **Class:** _____

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# Solids, liquids and gases: changing state

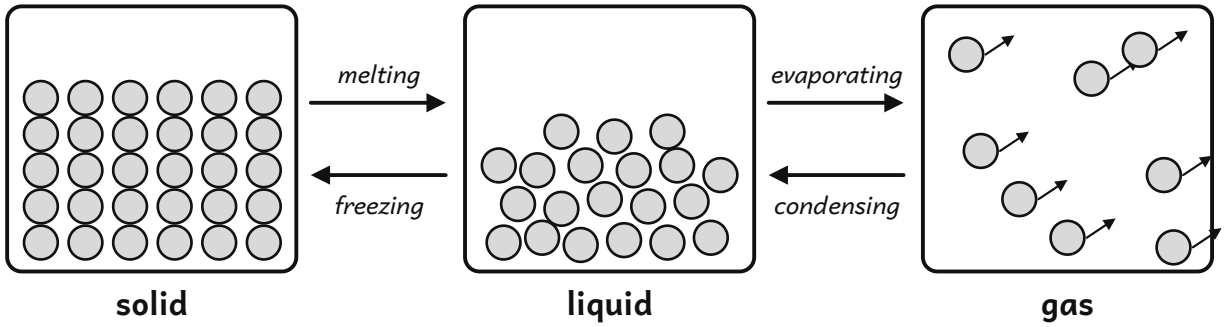
★☆☆ Grade 5

Name: Class: Date:

**Learning goals**

- Explain evaporation, condensation, boiling, melting and freezing.
- Recall that water boils at 100 °C and ice melts at 0 °C.
- Explain that adding or removing heat causes changes of state.

Wet clothes dry on the roof. A glass of cold water "sweats" on a hot day. Ice in a glass of lemon sherbet melts. A kettle whistles as water boils. All of these are **changes of state**, and every one is caused by **heating** or **cooling**.




*The particles in a solid, a liquid and a gas.*

## Three states of matter

In a **solid**, particles are packed tightly in rows and can only vibrate, so a solid keeps its shape. In a **liquid**, particles are close together but can move past each other, so a liquid flows and takes the shape of its container. In a **gas**, particles are far apart and move quickly in all directions, so a gas spreads out to fill any space.

## Evaporation

**Evaporation** is when a liquid turns into a gas. It happens at the **surface** of a liquid, at any temperature. Water evaporates faster when it is **warmer**, when it is **windy**, and when it has a **larger surface area**. That is why clothes dry fastest when they are spread out on a hot, windy day.

## Condensation

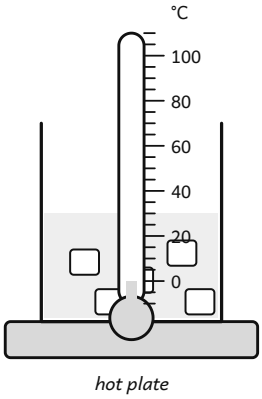
**Condensation** is the **reverse** of evaporation: a gas turns into a liquid when it **cools**. Air always contains some invisible **water vapour**. When it touches a cold surface, such as a cold glass, a window in winter or grass at night, it condenses into drops. This is how **dew** forms on winter mornings.

## Boiling

When water is heated strongly it **boils** at **100 °C**. Bubbles of water vapour form all through the liquid, not just at the surface. While it is boiling, the temperature stays at 100 °C: the heat energy is being used to turn liquid into gas.

## Melting and freezing

Ice **melts** at **0 °C**, and water **freezes** at 0 °C. Different materials melt at different temperatures: chocolate melts at about 30–35 °C (in your hand!), candle wax at about 60 °C, and iron at over 1500 °C.




*Heating ice and recording the temperature.*

## Evaporating a solution

If you dissolve salt in water and then let the water **evaporate**, the salt is left behind as crystals. The salt does not evaporate. This is how salt is made from sea water.

**From the past**

The **Celsius** scale was invented in 1742 by the Swedish astronomer Anders Celsius. Surprisingly, he first set 0 for boiling water and 100 for freezing water! It was turned the other way round after he died. In the USA people still often use the **Fahrenheit** scale, on which water freezes at 32 °F and boils at 212 °F.

**Did you know?**

On top of Mount Everest water boils at only about 70 °C, because the air pressure is much lower. It is hard to cook rice up there!

**Key words**

| | |
|---|---|
| **evaporation** | liquid → gas, at the surface, at any temperature |
| **condensation** | gas → liquid when it cools |
| **boiling** | liquid → gas all through the liquid, at the boiling point |
| **melting** | solid → liquid |
| **freezing** | liquid → solid |
| **water vapour** | water as a gas (invisible) |
| **melting point** | the temperature at which a solid melts |
| **boiling point** | the temperature at which a liquid boils |

FACT CHECK · C1 · STATES OF MATTER

# Changes of state

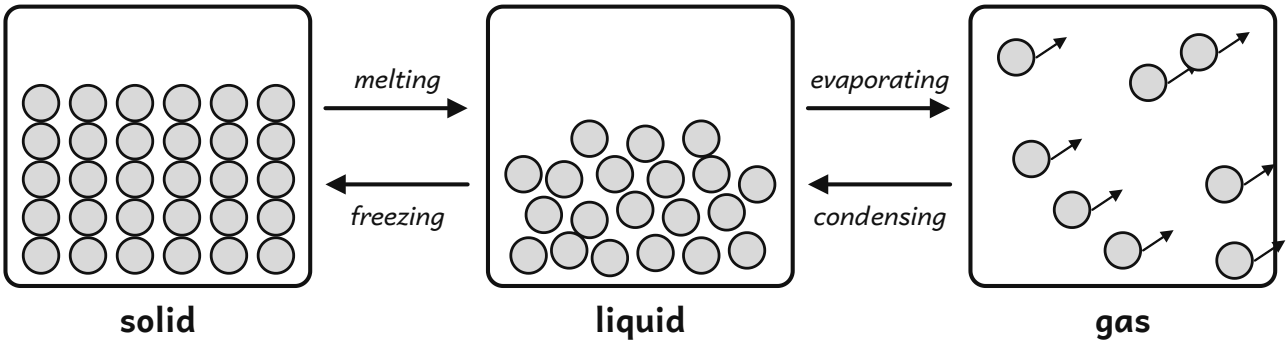
★☆☆
Grade 5

Name: Class: Date:

**Learning goals**

- Name the changes of state.
- Recall key temperatures.

**1** **Label the changes of state on the arrows.** [4]




| Change | Name |
|---|---|
| solid → liquid | |
| liquid → solid | |
| liquid → gas | |
| gas → liquid | |

**2** **Fill in the gaps.** [5]

WORD BANK

**100** **50** **evaporates** **condenses** **salt** **melts** **0**

Water boils at _____ °C. Ice melts at _____ °C. When water vapour touches a cold window it _____. Puddles dry up because the water _____. When salty water evaporates, the _____ is left behind.

**3** **Match each everyday event to the change of state.** [5]

Draw a line to match each pair, or write the letter.

| | |
|---|---|
| wet clothes drying ● | ● **A** melting |
| dew on the grass in the morning ● | ● **B** boiling |
| ice cream dripping ● | ● **C** freezing |
| a kettle making steam ● | ● **D** condensation |
| water turning into ice cubes in a freezer ● | ● **E** evaporation |

**4** **True or false?** [3]

| Statement | True | False | If false, correct it |
|---|---|---|---|
| 1. Evaporation only happens at 100 °C. | | | |
| 2. Condensation is the reverse of evaporation. | | | |
| 3. The temperature of boiling water keeps rising while it boils. | | | |

THINK IT THROUGH · C1 · STATES OF MATTER

# Explaining changes of state

★★☆
Grade 5

**Name:** **Class:** **Date:**

**Learning goals**

- Explain changes using heating and cooling.
- Interpret a heating graph.

**1** **Explain each observation.** [6]

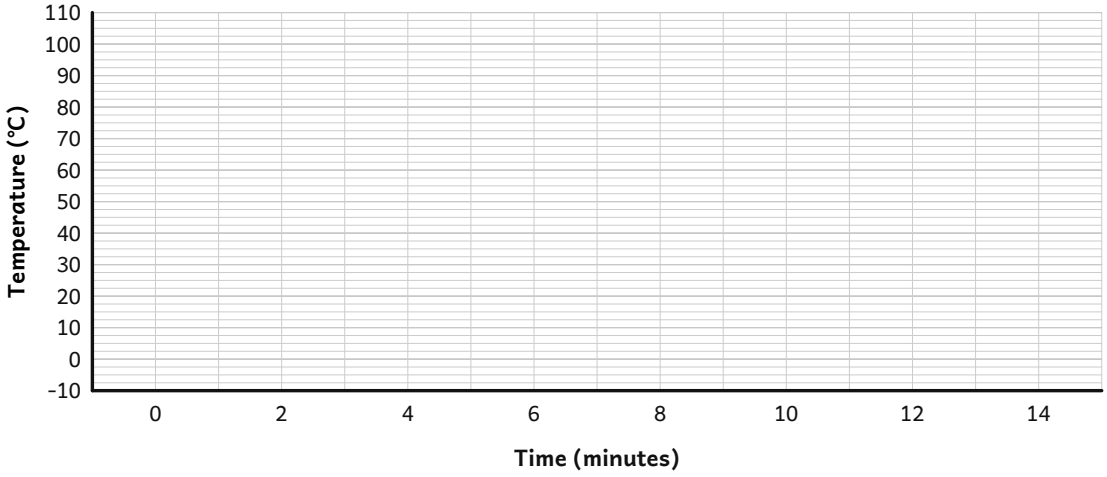
**a)** A cold bottle of water from the fridge becomes wet on the outside.

**b)** Clothes dry faster when they are spread out than when they are folded.

**c)** In winter, people's breath looks like a little cloud.

**2** **Class 5A heated ice and recorded the temperature every minute.** [6]

| Time (min) | 0 | 2 | 4 | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|---|---|---|
| Temp (°C) | −10 | 0 | 0 | 25 | 58 | 90 | 100 | 100 |




**a)** Plot the line graph.

**b)** Why did the temperature stay at 0 °C between 2 and 4 minutes?

.....

.....

**c)** What was happening at 12–14 minutes?

.....

.....

**3** **Why do people in Bangladesh store drinking water in clay pots (*kolshi*) to keep it cool?** [2]

.....

.....

.....

INVESTIGATE · C1 · STATES OF MATTER

# Does surface area affect evaporation?

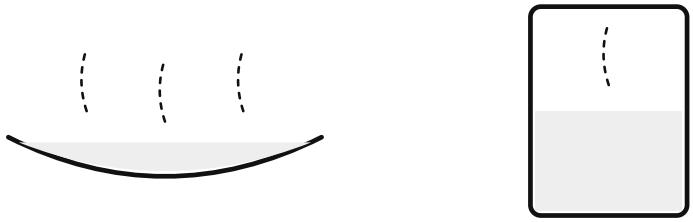
★★☆ Grade 5

**Name:** **Class:** **Date:**

**Learning goals**

- Plan a fair test.
- Plot two lines on one graph.
- Draw a conclusion.

*same volume of water: 100 ml*



**wide dish** **narrow cup**

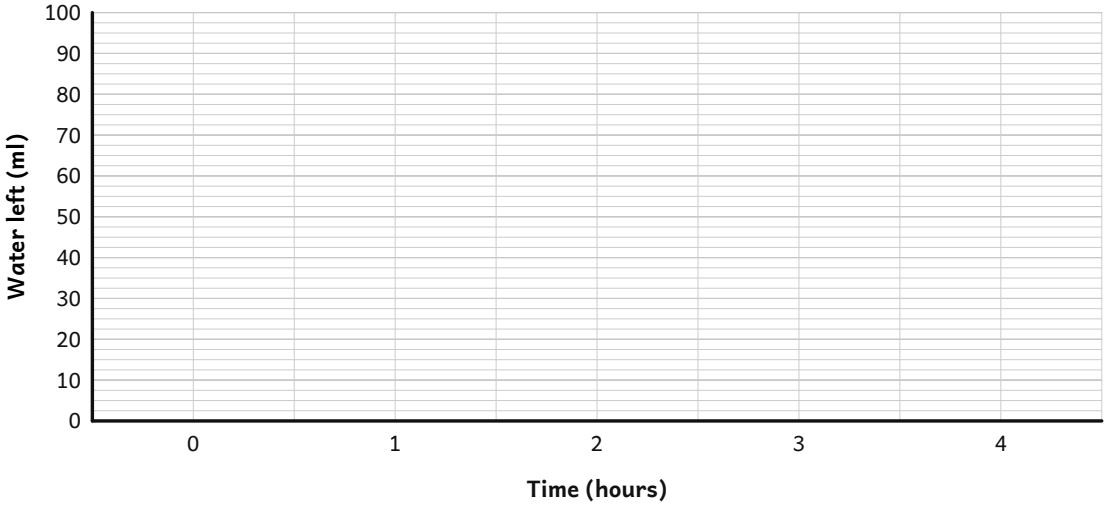
Arif and Mitu put 100 ml of water in a wide dish and 100 ml in a narrow cup. They left both in the same sunny place and measured the water left every hour.

**1** **Complete the plan.** [5]

| | |
|---|---|
| Question | |
| Change | |
| Measure | |
| Keep the same | |
| Prediction | |

**2** **Plot both sets of results on one line graph.** [5]

| Time (hours) | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Wide dish (ml) | 100 | 82 | 64 | 46 | 30 |
| Narrow cup (ml) | 100 | 96 | 92 | 89 | 85 |




**3** **How much water evaporated from each container in 4 hours?** [2]

.....

**4** **Write a conclusion.** [2]

.....

.....

# Salt from the sun: Cox's Bazar salt farmers

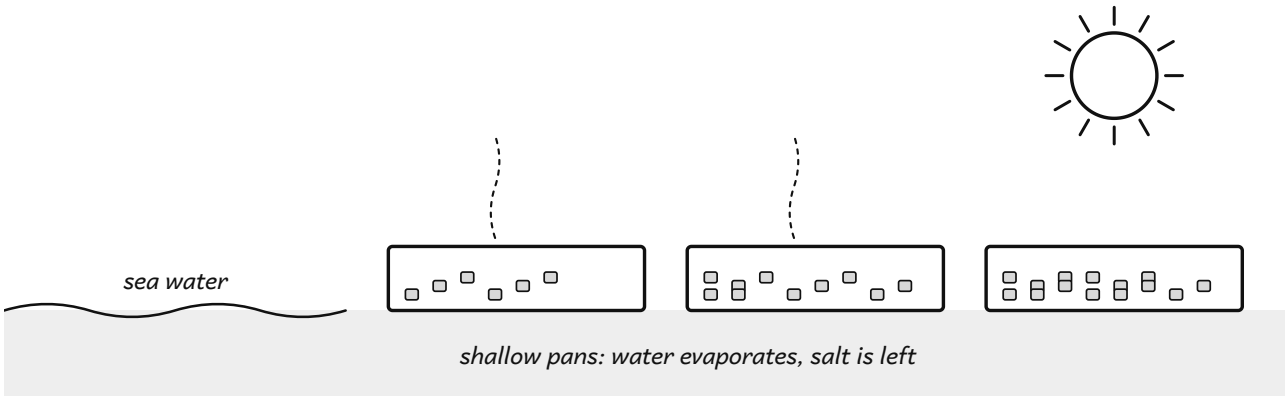
★★★ Grade 5

**Name:** **Class:** **Date:**

**Learning goals**

- Apply evaporation to a real industry.
- Use data and maths.
- Consider how weather and climate affect people.

Science · Geography · Maths · Economics · Health




*Salt pans: sea water is let into shallow fields and the Sun evaporates the water.*

Almost all the salt eaten in Bangladesh is made on the coast of **Cox's Bazar** and Chattogram. From about November to May, the dry season, farmers let sea water into shallow, flat fields covered with plastic sheets. The Sun and the wind **evaporate** the water, and white **salt crystals** are left behind. Farmers rake up the salt and carry it away in baskets.

In the monsoon, rain stops salt making: rain dissolves the salt and adds fresh water to the fields. Many salt farmers then grow shrimp or rice instead. Before it is sold, salt is cleaned and **iodine** is added, because our bodies need a small amount of iodine for healthy growth and brain development.

Based on descriptions of the Bangladesh salt industry (BSCIC and news reports). Numbers below are examples.

**1** **Explain how salt is made using the words *evaporate*, *dissolved* and *crystals*.** [3]

**2** **Why can salt not be made during the monsoon?** [2]

**3** **About 3.5 kg of salt is dissolved in every 100 litres of sea water.** [3]

**a)** How much salt is in 1000 litres of sea water?

**b)** A farmer's field holds 20 000 litres. How much salt could it make?

**4** **Why is iodine added to salt?** [1]

**5** **Rising sea levels and stronger storms may affect salt farms. Suggest one problem and one solution.** [2]

# Show what you know

★★★
Grade 5

Name: Class: Date:

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

**1** **Find the key words.**

| J | W | S | R | B | D | J | C | J | H |
|---|---|---|---|---|---|---|---|---|---|
| G | A | S | T | V | A | P | O | U | R |
| M | F | S | M | E | V | S | N | M | E |
| E | Y | O | Y | V | A | K | D | Y | B |
| L | F | L | L | H | J | M | E | M | H |
| T | R | I | P | I | H | G | N | N | E |
| U | E | D | F | P | Q | O | S | P | G |
| I | E | A | G | K | B | U | E | H | J |
| R | Z | I | B | M | E | I | I | T | W |
| V | E | W | U | B | O | I | L | D | L |

**boil condense freeze gas liquid melt solid steam vapour**

**2** **At what temperature does pure water boil?** [1]

- ◯ 0 °C
- ◯ 37 °C
- ◯ 100 °C
- ◯ 212 °C

**3** **Which change happens when dew forms?** [1]

- ◯ evaporation
- ◯ condensation
- ◯ melting
- ◯ boiling

**4** **Draw the particles in *a* solid, *a* liquid *and a* gas.** [3]

solid

liquid

gas

**5** **Challenge. Explain the difference between evaporation and boiling.** [2]

**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| describe solids, liquids and gases using particles | ○ | ○ | ○ |
| explain evaporation and condensation | ○ | ○ | ○ |
| recall that water boils at 100 °C and ice melts at 0 °C | ○ | ○ | ○ |
| explain why the temperature stays the same while ice melts | ○ | ○ | ○ |
| investigate what affects evaporation | ○ | ○ | ○ |
| explain how salt is made from sea water | ○ | ○ | ○ |